

Beneficial Effects of a Three-Month Structured Exercise Training Program on Cardiopulmonary Functional Capacity in Young Women with Polycystic Ovary Syndrome

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Context: Polycystic ovary syndrome (PCOS) is an endocrine disease closely related to several risk factors for cardiovascular disease. An impaired cardiopulmonary functional capacity was previously demonstrated in PCOS women. No data regarding the effects of a structured exercise training (ET) program on cardiopulmonary functional capacity in PCOS women are available.

Objective: Our objective was to evaluate the effects of a 3-month ET program on cardiopulmonary functional capacity in young PCOS women.

Design and Setting: A prospective baseline-randomized clinical study was conducted at the University “Federico II” of Naples, School of Medicine (Italy).

Patients: Ninety young overweight PCOS women were enrolled.

Mean Outcome Measures: Ninety young PCOS women were randomly subdivided into two groups, each composed of 45 subjects. The

PCOS-T (trained) group underwent a 3-month structured ET program, whereas the PCOS-UnT (untrained) group did not. Hormonal and metabolic profiles and cardiopulmonary and exercise parameters were evaluated.

Results: After 3-month ET, PCOS-T showed a significant improvement in peak oxygen consumption (+35.4%; $P < 0.001$) and in maximal workload (+37.2%; $P < 0.001$). In PCOS-T we also observed a significant reduction in body mass index (−4.5%; $P < 0.001$) and in C-reactive protein (−10%; $P < 0.001$), and a significant ($P < 0.001$) improvement in insulin sensitivity indexes. After 3 months, no changes were observed in PCOS-UnT.

Conclusions: A 3-month structured ET program improves cardiopulmonary functional capacity in young PCOS women. (*J Clin Endocrinol Metab* 92: 1379–1384, 2007)

POLYCYSTIC OVARY SYNDROME (PCOS) is an exceptionally common disorder affecting approximately 5–10% of reproductive-aged women (1). PCOS is the first cause of infertility in women, but it also should be considered a multifaceted disease (1–3). Chronic anovulation, hyperandrogenism, and insulin resistance (IR) are its main characteristics (4).

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Abbreviations: AUC, Area under the curve; AUC_{GLU} , AUC for glucose; AUC_{INS} , AUC for insulin; BMI, body mass index; CRP, C-reactive protein; CVD, cardiovascular disease; CVR, cardiovascular risk; DBP_{peak}, diastolic blood pressure at peak exercise; DBP_{rest}, diastolic blood pressure at rest; ET, exercise training; HR_{rest}, resting heart rate; IR, insulin resistance; LTPA, leisure-time physical activity; MET, metabolic equivalent; PCOS, polycystic ovary syndrome; PCOS-T, PCOS trained group; PCOS-UnT, PCOS untrained group; SBP_{rest}, systolic blood pressure at rest; VE/VCO_{2slope} , slope of increase in ventilation over carbon dioxide output; VO_{2AT} , oxygen consumption at anaerobic threshold; VO_{2max} , maximal oxygen consumption; $Watt_{max}$, maximal workload; WC, waist circumference; WHR, waist/hip ratio.

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IR, a well-recognized cardiovascular risk (CVR) factor (5), is frequently associated with other CVR factors (e.g. dyslipidemia, hypertension, glucose intolerance, and diabetes) (5), thus increasing CVR profile in young PCOS women (6, 7).

Treatment of PCOS is focused on both normalizing short-term signs of hyperandrogenism and anovulation and reducing metabolic complications (8).

Diet and exercise are recommended as first-line treatment of oligomenorrhea, hirsutism, infertility, and obesity in PCOS by the majority of endocrinologists and gynecologists (9). In PCOS women, short-term weight-loss intervention studies have decreased abdominal fat (10, 11), hyperandrogenemia (10), and improved insulin sensitivity (10, 11) and lipid profile (11). In non-PCOS subjects, lifestyle modification has also proved as efficacious as pharmacological intervention in reducing the risk of developing type 2 diabetes mellitus (12). The National Institutes of Health clinical guidelines for the long-term treatment of overweight and obesity emphasize the importance of achievable and sustainable goals, notably a combination of diet modification, physical activity, and behavior therapy (13).

In PCOS women, we recently documented an impaired cardiopulmonary functional capacity strictly related to IR (14). In women with proven coronary artery disease, an impaired cardiopulmonary functional capacity is associated with an increased risk of mortality for cardiovascular events (15, 16). Although nowadays PCOS women do not seem to be at increased risk for cardiovascular mortality (17), there are some intriguing reports showing higher CVR in PCOS even at an early age (6, 18).

Given the recognized beneficial effects of exercise training (ET) on oligomenorrhea, hirsutism, infertility, and obesity in PCOS women (9), this study was performed to establish whether these beneficial effects may be extended to cardiopulmonary functional capacity in young PCOS women.

Subjects and Methods

Subjects

Ninety consecutive young overweight nonsmoking women with PCOS were enrolled in the study protocol. All the PCOS patients achieved the European Society for Human Reproduction and Embryology/American Society for Reproductive Medicine criteria for the PCOS diagnosis (19). Polycystic ovaries were identified by transvaginal ultrasonography examination (20) and hirsutism by Ferriman-Gallwey score greater than 8 (21).

Exclusion criteria included pregnancy, glucose intolerance (as screened by a 2-h oral glucose tolerance test) and diabetes, hypothyroidism, hyperprolactinemia, Cushing's syndrome, nonclassical congenital adrenal hyperplasia, and use of oral contraceptives, glucocorticoids, antiandrogens, ovulation induction agents, antidiabetic or antiobesity drugs, or other hormonal drugs within the previous 6 months. Subjects with neoplastic, hepatic, respiratory, and any cardiovascular disorder or other concurrent medical illness (*i.e.* heart failure, lung or renal disease) were also excluded from the study. None of the study patients drank alcoholic beverages.

Protocol and treatment

The study was conducted according to the guidelines of the Declaration of Helsinki, and the institutional ethical committee approved the study protocol. The purpose of the protocol was explained to each subject, and written informed consent was obtained from each patient before beginning the study. The study had no external funding source.

A common core of assessments was performed at baseline and after 3 months. All patients underwent blood sampling for a hormonal assessment, lipid profile, and fasting glucose and insulin levels (see *Biochemical assays*). During the same visit, all subjects underwent cardiovascular and endocrinological examination, 12-lead electrocardiography, transvaginal ultrasonography, cardiopulmonary exercise test, anthropometric measurements, including height, weight, body mass index (BMI; ratio between the weight and the square of the height), waist circumference (WC; measured at the midpoint between the lateral iliac crest and the lowest rib margin at the end of normal expiration), waist/hip ratio (WHR; ratio between the smallest circumference at the torso and the widest circumference at the hip), and leisure-time physical activity (LTPA) questionnaire (22) as detailed below. At study entry, general dietary and behavioral advice without a structured caloric restriction program was given to the entire PCOS study population. All of the PCOS population was counseled to achieve a healthy balanced meal plan with regular food with a nutritional composition of 50% of calories from carbohydrate, 25% from protein, and 25% from fat. Intake of low glycemic index foods was encouraged.

At study entry, PCOS women were randomly subdivided into two groups composed of 45 patients each; the PCOS-T (trained) group underwent a 3-month structured ET program, whereas the PCOS-UnT (untrained) group did not. None of the subjects received any medications throughout the study.

All clinical assessments were performed by the same physician who was blinded to the patient allocation into the study protocol. All study

procedures were completed under the same conditions, and at the same time of day for both the baseline and follow-up tests.

Each patient was instructed to have a personal daily diary for noting the characteristics of the menses (duration and frequency) and the occurrence of serious events. This personal diary was checked after 3 months from starting ET.

Biochemical assays

All blood samples were obtained in the morning between 0800 and 0900 h after an overnight fasting during the early follicular phase (d 2–4) of progesterone-induced menstrual cycle. Blood samples were collected into tubes containing EDTA after a 30-min resting period in the supine position. All blood samples were immediately centrifuged at 4°C for 20 min at $1600 \times g$ and stored at -20°C until assayed.

Plasma LH, FSH, prolactin, estradiol, progesterone, 17α -hydroxyprogesterone, testosterone, androstenedione, and dehydroepiandrosterone sulfate levels were measured by specific RIAs as previously described (5, 6). The levels of SHBG were measured using an Immunoradiometric assay (5, 6), and the free androgen index was calculated [testosterone (nmol/liter)/SHBG (nmol/liter) $\times 100$]. Blood insulin and glucose levels were measured by a solid-phase chemiluminescent enzyme immunoassay and the glucose oxidase method, respectively (5, 6).

The glucose and insulin areas under curve (AUCs) and the AUC glucose (AUC_{GLU})/AUC insulin (AUC_{INS}) ratio (23), in response to the oral glucose tolerance test, were also calculated. The lipid profile consisted of serum total cholesterol, high-density lipoprotein-cholesterol, low-density lipoprotein-cholesterol, and triglyceride levels as previously described (24). C-reactive protein (CRP) was measured as recently shown (25).

Cardiopulmonary exercise test and exercise training program

All patients underwent an incremental cardiopulmonary exercise test on a bicycle ergometer as previously described (14).

The PCOS-T group underwent a 3-month structured ET program on a hospital ambulatory-based regimen. Training sessions, performed three times per week under continuous electrocardiographic monitoring, were supervised by a cardiologist, a physiotherapist, and a graduate nurse. Each session was preceded by a 5-min warm-up and followed by a 5-min cool-down. Exercise was performed for 30 min on a bicycle ergometer with the target of 60–70% of the maximal oxygen consumption ($\text{VO}_{2\text{max}}$) achieved at the initial cardiopulmonary exercise test monitored by a wearable device. Exercise workload was gradually increased until the achievement of the predefined target.

LTPA questionnaire

The self-reported LTPA, including all recreational activities, housework, and yardwork (*e.g.* sweeping, shopping, gardening, walking, different types of sports, *etc.*) was recorded. Using a standardized classification of the energy expenditure associated with physical activities (26), we calculated a weekly energy expenditure score (total LTPA level) in metabolic equivalents per hour per week (METs-h/wk). LTPA level was graded into four categories of increasing order with the aid of the following scheme: 1) no weekly LTPA; 2) only light LTPA most of the week; 3) strenuous LTPA (large increase in heart rate, breathing, and perspiration) for at least 20 min once or twice a week; and 4) strenuous LTPA for at least 20 min three times a week or more. To simplify this gradation, we shortened the description into no, low, moderate, and high LTPA. Occupational physical activity was not included in the analysis because of the subjectivity of the questions and its different nature compared with the question about physical activity at home. Finally, participants were asked to estimate the number of hours per day they spent engaged in sedentary behavior, including time spent sitting as well as lying down or sleeping.

Statistical analysis

For categorical variables the χ^2 test was performed. For two related categorical variables, the McNemar-Bowker test of symmetry was used.

For continuous variables, the unpaired- and paired-samples *t* tests

were used to compare the means of the two groups of cases and the means of two variables (baseline and third month) in each group, respectively. The bivariate correlations procedure was used to compute Pearson's correlation coefficients with the significance levels.

For all subjects, multiple linear regression analysis (stepwise method) was performed with VO_{2max} at 3 months as dependent variable and with baseline data [AUC_{INS} , AUC_{GLU}/AUC_{INS} , age, BMI, resting heart rate (HRrest), systolic blood pressure at rest (SBPrest), diastolic blood pressure at rest (DBPrest), and membership in the untreated or the treated group (code 0/1)] as independent variables. In assessing the suitability of the data for a linear regression model, the colinearity diagnostics were evaluated. Two-sided *P* values of less than 0.05 were considered to indicate statistical significance. Statistical tests were performed with SPSS software, version 15.0.0 (SPSS Inc., Chicago, IL).

Results

Baseline anthropometric and hormonal profiles of the study population are summarized in Table 1. All PCOS patients showed polycystic ovaries and anovulation. Seventy-nine patients (87.8%) had clinical hyperandrogenism, and 27 (63.3%) showed biochemical hyperandrogenism.

All subjects completed the study protocol. The duration of the intervention period was 84 ± 3 d in the PCOS-T group. No adverse events took place during any of the training sessions. In PCOS-T, the average exercise intensity was $67 \pm 4\%$ of initial VO_{2max} , and averaged 92 ± 28 min per week of training session. Attendance was 100% in the PCOS-T group for the 3-month ET program (28 ± 2 training sessions). After intervention, 27 of 45 (60%) women of the PCOS-T group showed normal menstrual cycles.

Metabolic, CVR profile, cardiopulmonary and hemodynamic parameters are summarized in Table 2. Only in PCOS-T patients after 3 months of ET was there a significant improvement of BMI, WC, WHR (Table 1), fasting insulin, and AUC_{INS} from baseline (Table 2). Moreover, in PCOS-T after 3 month of ET, BMI ($P < 0.05$), WC ($P < 0.01$), WHR ($P < 0.05$), fasting insulin ($P < 0.01$), and AUC_{INS} ($P < 0.001$) were significantly different when compared with PCOS-UnT (Tables 1 and 2). Furthermore, only in PCOS-T was the AUC_{GLU}/AUC_{INS} ratio significantly ($P < 0.001$) increased compared with baseline, with a difference ($P < 0.001$) between PCOS-T and PCOS-UnT at 3 months (Table 2).

In PCOS-T, after 3-month ET, the improvement in VO_{2max}

was significantly correlated to the BMI reduction ($r = -0.605$; $P < 0.001$), WC reduction ($r = -0.561$; $P < 0.01$), WHR reduction ($r = -0.524$; $P < 0.01$), to the AUC_{INS} reduction ($r = -0.452$; $P < 0.01$); and to the increase in AUC_{GLU}/AUC_{INS} ratio ($r = 0.486$; $P < 0.01$).

Free androgen index (Table 1) and lipid profile (Table 2) were unchanged after 3 months of follow-up in both the PCOS-T and PCOS-UnT groups. Baseline values of CRP were higher than normal (normal values < 1 mg/ml) in both groups but decreased significantly after ET only in PCOS-T (Table 2).

There were no significant differences between the two groups in baseline hemodynamic and cardiopulmonary parameters (Table 2).

After a 3-month structured ET program, in PCOS-T a significant improvement was observed in VO_{2max} , oxygen consumption at anaerobic threshold (VO_{2AT}), maximal workload ($Watt_{max}$), HRrest, and SBPrest (Table 2), whereas no significant change was detected in PCOS-UnT. After a 3-month structured ET program, a significant improvement in VO_{2max} ($P < 0.001$), VO_{2AT} ($P < 0.001$), and $Watt_{max}$ ($P < 0.001$) was observed in PCOS-T compared with PCOS-UnT (Table 2). A significant reduction in the slope of increase in ventilation over carbon dioxide output (VE/VCO_{2slope}) ($P < 0.05$), HRrest ($P < 0.001$), SBPrest ($P < 0.01$), and diastolic blood pressure at peak exercise (DBPpeak) ($P < 0.01$) was also observed in PCOS-T compared with PCOS-UnT (Table 2).

At entry, no difference between PCOS-T and PCOS-UnT patients reporting no or low LTPA level was observed (64 vs. 66%, PCOS-T vs. PCOS-UnT patients, respectively) (Table 3). At 3-month follow-up, a significant improvement in LTPA level was observed in PCOS-T ($P < 0.001$); conversely, in PCOS-UnT, LTPA level remained unchanged (Table 3).

In PCOS-T patients, after 3-month ET program, total LTPA level was significantly ($P < 0.001$) increased, whereas no statistically significant changes in PCOS-UnT were observed (Table 3). In PCOS-T patients, 3-month changes in total LTPA level were significantly correlated with changes in VO_{2max}

TABLE 1. Baseline and 3-month anthropometric and hormonal profile in PCOS-T and PCOS-UnT study population

	PCOS-T (n = 45)			PCOS-UnT (n = 45)		
	Baseline	Third month	<i>P</i>	Baseline	Third month	<i>P</i>
Age (yr)	21.7 ± 2.3			21.9 ± 1.9		
BMI (kg/m^2)	29.3 ± 2.9	28.0 ± 2.9	<0.001	29.4 ± 3.5	29.3 ± 3.2	0.88
WC (cm)	94.5 ± 3.4	91.8 ± 3.6	<0.001	94.0 ± 3.0	93.8 ± 3.1	0.87
WHR	0.84 ± 0.1	0.80 ± 0.1	<0.001	0.86 ± 0.1	0.85 ± 0.1	0.63
Ferriman-Gallwey score	11.9 ± 3.5	11.5 ± 3.1	0.56	12.1 ± 3.4	12.3 ± 3.2	0.77
Hemoglobin (g/dl)	13.5 ± 1.2	13.2 ± 1.4	0.27	13.6 ± 1.1	13.4 ± 1.3	0.43
FSH (IU/liter)	10.5 ± 1.7	10.3 ± 1.3	0.53	10.1 ± 1.6	10.2 ± 1.2	0.73
LH (IU/liter)	24.2 ± 3.3	23.5 ± 3.2	0.31	23.5 ± 3.1	23.1 ± 2.9	0.53
Prolactin (ng/ml)	10.5 ± 1.3	10.3 ± 1.1	0.43	10.2 ± 1.1	10.1 ± 1.2	0.68
Estradiol (pmol/liter)	120 ± 30.5	115 ± 32	0.45	118 ± 26.1	116 ± 29	0.73
Progesterone (nmol/liter)	1.2 ± 0.4	1.1 ± 0.3	0.18	1.3 ± 0.6	1.2 ± 0.5	0.39
17 α -hydroxyprogesterone (nmol/liter)	1.6 ± 0.3	1.5 ± 0.5	0.25	1.5 ± 0.4	1.6 ± 0.3	0.18
Testosterone (nmol/liter)	2.3 ± 0.7	2.1 ± 0.6	0.14	2.5 ± 0.5	2.4 ± 0.4	0.29
Androstenedione (nmol/liter)	5.1 ± 0.7	4.9 ± 0.8	0.21	5.3 ± 0.9	5.1 ± 0.7	0.24
Dehydroepiandrosterone sulfate (μ mol/liter)	4320 ± 465	4168 ± 512	0.14	4290 ± 441	4175 ± 395	0.19
SHBG (nmol/liter)	27 ± 6.2	29 ± 5.8	0.11	29 ± 6.5	28 ± 6.9	0.48
Free androgen index	8.5 ± 3.4	8.7 ± 3.4	0.43	8.6 ± 3.6	8.5 ± 3.5	0.94

TABLE 2. Baseline and 3-month metabolic, cardiovascular risk profile, cardiopulmonary, and hemodynamic parameters in PCOS-T and PCOS-UnT study population

	PCOS-T (n = 45)			PCOS-UnT (n = 45)		
	Baseline	Third month	P	Baseline	Third month	P value
Fasting glucose (mg/dl)	94.9 ± 6.8	93.8 ± 5.5	0.16	95.3 ± 7.1	95.2 ± 7.2	0.93
Fasting insulin (μU/ml)	20.1 ± 3.5	18.3 ± 3.0	<0.001	20.2 ± 3.8	20.4 ± 3.6	0.80
AUC _{GLU}	12,329 ± 1,987	12,359 ± 2,230	0.85	12,020 ± 2,115	12,363 ± 2,228	0.32
AUC _{INS}	16,520 ± 949	14,750 ± 992	<0.001	16,476 ± 933	16,455 ± 985	0.52
AUC _{GLU} /AUC _{INS} ratio	0.73 ± 0.2	0.80 ± 0.1	<0.001	0.73 ± 0.2	0.72 ± 0.2	0.73
Total cholesterol (mg/dl)	153.0 ± 18.1	151 ± 16.4	0.34	154 ± 16	156 ± 17	0.47
LDL-C (mg/dl)	76.3 ± 18.3	73.2 ± 23.8	0.28	72.5 ± 21.6	75.9 ± 24.9	0.50
HDL-C (mg/dl)	53.8 ± 11.6	55.2 ± 12.9	0.50	59 ± 19	58 ± 16	0.74
Triglyceride (mg/dl)	114.6 ± 20.3	113.3 ± 23.3	0.68	112 ± 21	111 ± 23	0.84
CRP (mg/liter)	1.85 ± 0.7	1.54 ± 0.5	<0.001	1.77 ± 0.8	1.87 ± 0.9	0.58
VO _{2max} (ml/kg/min)	17.6 ± 2.5	23.7 ± 2.6	<0.001	17.7 ± 2.2	17.9 ± 1.8	0.37
VO _{2AT} (ml/kg/min)	13.7 ± 2.1	18.1 ± 2.7	<0.001	13.9 ± 2.0	14.0 ± 1.2	0.52
VE/VCO _{2slope}	27.1 ± 2.1	26.5 ± 2.3	0.13	27.2 ± 2.3	27.5 ± 1.8	0.56
RER	1.15 ± 0.03	1.16 ± 0.04	0.01	1.14 ± 0.04	1.15 ± 0.05	0.49
Watt _{max} (watts)	101.5 ± 20.1	133.8 ± 9.8	<0.001	100.4 ± 18.7	99.8 ± 13.1	0.84
HR _{rest} (beats/min)	77.0 ± 5.5	74.8 ± 4.8	0.03	77.2 ± 4.5	78.1 ± 3.9	0.31
HR _{peak} (beats/min)	147.7 ± 14.5	145.0 ± 11.0	0.27	146.9 ± 13.4	147.8 ± 13.9	0.77
SBP _{rest} (mm Hg)	117.9 ± 10.2	114.1 ± 10.6	0.02	118.7 ± 8.9	119.8 ± 9.5	0.55
SBP _{peak} (mm Hg)	167.3 ± 8.3	166.0 ± 8.7	0.45	166.8 ± 8.1	168.4 ± 7.5	0.33
DBP _{rest} (mm Hg)	76.1 ± 4.7	74.4 ± 5.0	0.07	75.8 ± 4.6	75.5 ± 3.9	0.59
DBP _{peak} (mm Hg)	84.9 ± 4.5	84.1 ± 3.6	0.33	85.5 ± 4.0	86.1 ± 3.4	0.39

LDL-C, Low-density lipoprotein cholesterol; HDL-C, high-density lipoprotein cholesterol; RER, respiratory exchange ratio; HR_{peak}, heart rate at peak exercise; SBP_{peak}, systolic blood pressure at peak exercise.

($r = 0.502$; $P < 0.001$), AUC_{INS} ($r = -0.417$; $P < 0.01$), and AUC_{GLU}/AUC_{INS} ratio ($r = 0.441$; $P < 0.01$).

The final model of multiple linear regression analysis showed that only the inclusion into the PCOS-T group was linearly related to higher VO_{2max} values at 3 months (unstandardized coefficient ± SE, 5.773 ± 0.479 ; standardized coefficient (β) 0.789 ; $P < 0.001$; constant 17.924).

Discussion

PCOS women represent an intriguing biological model illustrating the relationship between hormonal profile and CVR. Although the epidemiology is inconclusive, recent surrogate endpoint studies strongly supported an association between PCOS and cardiovascular disease (CVD) (27).

Discrepancies among studies may be due to small sample sizes, bias in case-control designs, and use of nonstandard definitions of PCOS (27). The evidence, based solely on association studies to date, indicates that IR and obesity may be the mediators of early ventricular abnormalities (6), en-

dothelial dysfunction (7), and both carotid (7) and coronary atherosclerosis (18) in PCOS. The mechanisms appear likely to be related to the consequences of IR.

Skeletal muscle is the major site of insulin-mediated glucose disposal and is implicated in the pathogenesis of IR (28). Some reports (29, 30) suggest that insulin action may be related to the oxidative capacity of skeletal muscle. VO_{2max} represents a validated index for assessing cardiovascular functional capacity (31). The levels of physical activity and insulin sensitivity are positively correlated (32), and VO_{2max} is considered a strong determinant of the insulin sensitivity index in both men and women (33). The increased prevalence of IR in PCOS and its strong association with inactivity may lead to an exercise-deficient phenotype (34), manifested by a greater disruption both of skeletal and muscle cellular metabolism and also correlated to a chronic low-grade inflammatory state (35).

We recently described a reduced cardiopulmonary functional capacity in a well-selected sample of PCOS patients in

TABLE 3. Baseline and 3-month LTPA level in PCOS-T and PCOS-UnT study population

	PCOS-T (n = 45) (%)			PCOS-UnT (n = 45) (%)		
	Baseline	Third month	P	Baseline	Third month	P
LTPA			<0.001 ^a			0.11 ^a
No	20 (44)	9 (20)		19 (42)	21 (47)	
Low	9 (20)	10 (22)		11 (24)	11 (24)	
Moderate	9 (20)	16 (36)		9 (20)	8 (18)	
High	7 (16)	10 (22)		6 (14)	5 (11)	
Total LTPA level (METs-h/wk)	8.6 ± 5.3	12.2 ± 5.2	<0.001 ^b	8.4 ± 6.7	8.3 ± 5.3	0.81 ^b

Data are n (%) except where indicated otherwise. At baseline PCOS-T vs. PCOS-UnT, $P = 0.78$ by the χ^2 test. At third month PCOS-T vs. PCOS-UnT, $P = 0.03$ by the χ^2 test.

^a By McNemar-Bowker test.

^b By the paired-samples t test.

comparison to age-matched healthy young women (14). Similar results were subsequently confirmed in a wider PCOS population (36).

Recently, in a small sample size obese PCOS population (12 patients), endurance and resistance exercise plus nutritional counseling (seven patients) resulted in a significant improvement of estimated oxygen consumption and in a decreased body fatness compared with PCOS women enrolled in the nutritional counseling group alone (five patients), whereas no statistical differences in hormonal profile (with the exception of insulin levels) and with no weight loss were observed in the entire PCOS study group (37). However, the decrease in body fatness (as shown by the significant decrease in the sum of two skin folds) with no weight reduction in trained obese PCOS women ($BMI = 36.2 \pm 2.0$) has to be interpreted in the absence of data regarding LTPA level, also taking into consideration the small sample size population (37).

In the present study, we demonstrate that a 3-month structured ET program induced a significant improvement in cardiopulmonary functional capacity and insulin sensitivity and also led to a significant reduction in BMI and CRP levels in a wide overweight PCOS population, named PCOS-T. In our view, the significant BMI and WC reduction observed in our PCOS-T group represent the combined effect of the complete adherence to the ET program sessions and of the significant increase in LTPA level.

Combined IR and exercise-deficient phenotype resulted in the present studied PCOS women, providing an unfavorable CVR profile characterized by increased BMI, low maximal cardiopulmonary functional capacity, reduced physical fitness, and low-grade chronic inflammatory state.

It is well known that exercise improves glucose homeostasis related to up-regulation of the expression and/or activity of proteins involved in insulin signal transduction in skeletal muscle (38). Moreover, inflammatory markers have been linked to the risk of CVD (36), and several reports (39–41) have shown that ET reduces CRP in healthy subjects (40) and that regular physical activity suppresses inflammation by reducing body adiposity and improving insulin sensitivity (41, 42).

Physical fitness is associated with a lower risk of CVD (43), early onset of metabolic syndrome (44), and a reduced mortality (15). In women referred for cardiac rehabilitation, Kavanagh *et al.* (16) reported that oxygen consumption was a strong independent predictor of cardiac mortality and that each increase of 1 ml/kg·min of initial peak oxygen consumption was associated with a 10% lower cardiac mortality.

Despite the presence of these CVR factors in PCOS women, there are no adequate prospective data documenting the effects of a structured ET program in PCOS women.

The current prospective randomized study was designed for evaluating the effects of a structured aerobic ET program on cardiopulmonary functional capacity in a wide PCOS population with no exercising habit (no or low LTPA level). Thus, the exercise-induced increase in cardiopulmonary functional capacity, the increased LTPA level, the BMI and WC reduction, the improvement of IR, and the reduced low-grade chronic inflammatory state may play a role conferring

to the ET structured program a significant long-term prognostic advantage.

Further studies will be necessary to extend our results to a larger and older population of women with PCOS, and long-term follow-up studies are required to evaluate whether these findings may translate into long-term reduction of CVR.

In conclusion, a structured 3-month ET program improved cardiopulmonary functional capacity and physical fitness, insulin sensitivity, and BMI, and reduced CRP in young overweight PCOS women. Given the strong evidence for a direct role of physical activity in the prevention of IR and the fact that ET increases mitochondrial biogenesis and improves glucose tolerance and insulin action in IR subjects, the present work strengthens the recommendation to apply primary defense mechanisms such as exercise in young women with PCOS. A structured ET program represents a simple therapeutic option that can be safely, routinely, and extensively performed for reducing CVR profile in young PCOS women.

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